

1.3a

The Neuron and Neural Firing Neural communication & the Endocrine System

What are Neurons & how do they transmit info?

Neurons are the building block of our body's neural info system.

Throughout life, new neurons are born & old neurons wither away.

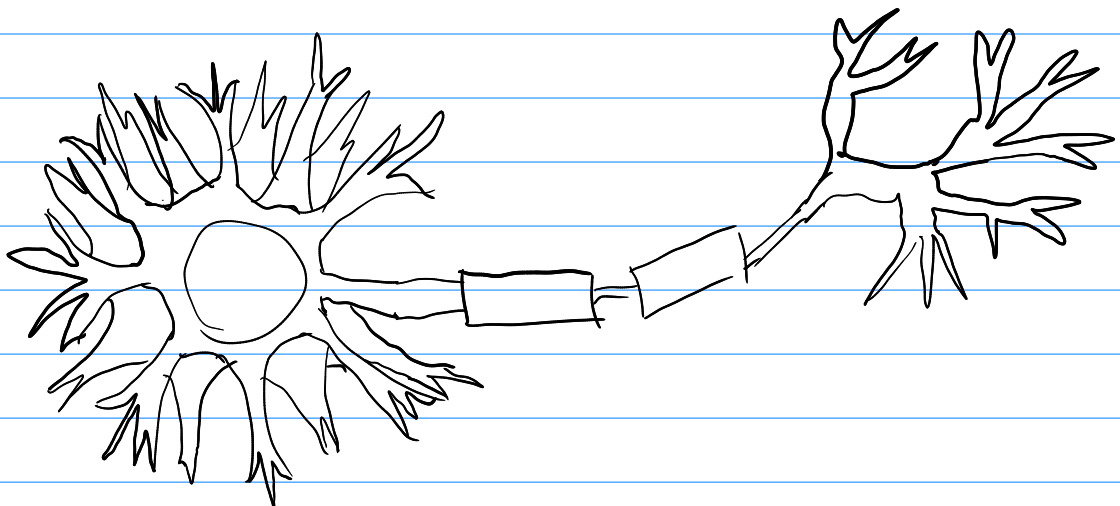
A neuron is composed of:

a cell body.

Bushy dendrites, fibers that receive & integrate info, conducting it → cell body.

A single long axon fiber passes the message through its terminal branches to other neurons or to muscles or glands.

Dendrites listen; Axons speak.



Unlike the short dendrites, axons are very long. Certain cells, incl. cells that carry signals CNS $\rightarrow$ muscles, can have very long axons, sometimes reaching several feet.

Some axons are insulated with a myelin sheath, a layer of fatty tissue.

MS speeds impulses. As MS is laid down while (age $\lesssim$ 25):

natural efficiency, judgement, self control $\uparrow$

MS degenerates $\Rightarrow$ control $\downarrow$, reaction time $\uparrow$

in extreme cases, causes multiple sclerosis:

communication to muscles and brain regions slow

$\therefore$ muscle control $\downarrow$, sometimes cognition $\downarrow$

Glia cells support our billions of neurons

Glia are the worker bees to the Neuron's Queen Bee

They provide nutrients & myelin, guide neural connections, clean up waste from neurons sending messages

By 'chatting' with neurons, they participate in info transfer.

complexity (brain) $\uparrow \Rightarrow$ glia: neurons $\uparrow$

brain (Einstein), # neurons $\nearrow$ average
concentrations (glial cells) $\nearrow$ average

Stimulation $\triangleleft$ neurons, senses
 $\Rightarrow$ neuron sends messages: firing impulse $\therefore$ action potential:
 $\checkmark$ + electrical charge travels $\downarrow$ axon

$\therefore$ fiber type $\Rightarrow$ Speed (impulse)
 $\sim 3 \frac{\text{km}}{\text{h}} \leq \text{speed} \leq 320 \frac{\text{km}}{\text{h}}$, $\text{max}(\text{speed neuron}) \approx 3 \text{ mil} = \text{speed}(\text{electricity in wire})$

Brain activity, Δ ms (10^{-3} s)
Computer activity, Δ ns (10^{-9} s)

Response (computer) ≈ 0 , but your sudden response ≥ 0.2 s

Complex (brain) $\gg \gg$ complexity (computer)
speed (brain) $\ll \ll$ speed (computer)

w/ (chemical events, $\sim$ to batteries), neurons $\triangleright$ electricity

ions exchanged; fluid $\Leftarrow$ axon membrane has + charge
fluid $\Rightarrow$ resting axon, $\sim$ - charge

Resting Potential: $\Leftarrow$ is +, $\Rightarrow$ is -

neuron fires { section 1 (axon) opens gates
+ ions (attracted to $- \oplus$) come in
loss of Δ charge :: depolarization
 $\Rightarrow$ next section of axon opens gates

temp inflow + charge :: neural impulse :: action potential

neuron $\triangleright$ complex calculations 100-1000 times

↑
signals from 100s to 1000s $\bigcirc$ neurons

neuron tells $\bigcirc$ neuron $|v|$ (excited)

most nu signals = excitatory

$\bigcirc$ = inhibitory

$\sum \text{excited} + \text{threshold} > \sum \text{inhibited}$

$\Rightarrow$ signals $\rightarrow$ action potential

$\rightarrow$ action potential travels $\downarrow$ axon,

branches into junctions w/ 100s-1000s $\bigcirc$ neurons
 $\vee$ glands, muscles

while (resting pause :: refractory period): no action potential,
until axon recharges, resting state $\Rightarrow$ fire again

$|v| > \text{threshold}$ ($|v|$ (stimulation) $\uparrow$ $\nRightarrow$ intensity (nu impulse)

impulse = all-or-none-response

$|v|$ (stimulus) $\Rightarrow$ # neurons firing, speed (firing) $\uparrow$

How do neurons communicate?

neurons very interwoven

very small gap axons $\longleftrightarrow$ dendrites :: synapse

action potential reaches button-like terminal at end of axon
 $\Rightarrow$ chem. messengers :: neurotransmitters released

Within 10^{-4} s, NT molecules cross synaptic gap
bind to receptor sites precisely.

for instance, NT unlocks tiny channels

ions flow in $\Rightarrow$ exciting or inhibiting receiving neuron
enzymes break down, sending neuron reabsorbs excess NTs
:: reuptake.

Some antidepressants block reuptake of mood $\uparrow$ NTs

How do NTs influence us?

Brain Pathway $\propto$ may use 1, 2 NTs

NTs (depend on brain location) affect behaviour, emotion

NT systems don't work alone: interact
receptors stimulate $\rightarrow$ effectors

Brain makes opioid-like NTs :: endorphins $\rightarrow$ control pain

Flooding brain w/ artificial opioids would
disrupt brain's chem. balancing
brain may stop Δ endorphins

$\therefore$ Agonist molecules $\Rightarrow$ NT's action $\uparrow$

$\therefore$ Some Agonists $\rightarrow$ production, release (NT) $\uparrow$

○ Agonists block reuptake

○ Agonists $\approx$ NT $\therefore$ bind to its receptors
mimicking it

$\therefore$ Antagonists $\rightarrow$ NT's action $\downarrow$
by blocking production & release.

Some Antagonists are similar enough to a
NT to occupy its receptor, but not
similar enough to stimulate it.

What is the Endocrine System

Endocrine System is interconnected with
Nervous System.

EndSys secretes other chem. messengers $\therefore$ hormones

Some hormones chem. $\equiv$ neurotransmitter

NerSys very fast, EndSys takes $\geq$ several sec to
go from gland to target tissue

End. messages **outlast** effect of **nu** messages.

In danger, $\therefore$ ANS $\rightarrow$ adrenal glands $\triangleright$ adrenaline & nor adrenaline; hormones $\rightarrow$ HR, BP, BS $\uparrow$
After this, the shaken up feeling lingers

Hypothalamus controls Pituitary Gland, the most influential end. gland.

Brain $\rightarrow$ Pituitary $\rightarrow$ other glands $\rightarrow$ hormones $\rightarrow$ body & brain

MCQ: 1c, 2a, 3d, 4d, 5d, 6c
7b, 8b, 9d

Personal Response:

The concept of **nu** connections is one I've taken into account throughout my life. It was the primary motivator for me to try and sleep more, as more **nu** connections can develop while sleeping.